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Addressing Challenging Well Conditions: Optimized Cementing Approaches for Achieving Zonal Isolation in Intermediate and Deep Liners

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Abstract

Addressing challenging well conditions to achieve zonal isolation during primary cementing operations is a top priority for ensuring cementing success, well integrity, and maintaining safety, as well as regulatory and environmental compliance. This case study presents two successful field applications of liner cementing operations conducted within a narrow operating window (between formation pore and fracture pressures) across challenging formations.

To address these conditions, a managed pressure cementing (MPC) technique was implemented to overcome challenges posed by the narrow operating window. The approach also incorporated engineered heavy-weight slurries, a wellbore shielding spacer combined with an environmentally friendly lost circulation material (LCM), and an expandable liner hanger system. Extensive placement simulations and meticulously designed fluid properties helped optimize cement placement accuracy. The case studies involved 9 5/8-in and 7-in liners, where the deviated sections were drilled using the Managed Pressure Drilling (MPD) technique from the preceding 13 3/8-in and 9 5/8-in casings, respectively, through weak formations intersecting high-pressure zones.

Although the well conditions were complex, zonal isolation was achieved in both cases by incorporating pioneering cementing techniques. The use of MPC, along with wellbore shielding spacer system helped mitigate potential losses within the narrow operating window and ensured effective zonal isolation. Extensive placement simulations contributed to maintaining well stability throughout the execution. The use of expandable liner hanger systems, combined with detailed engineering procedures and meticulously designed fluid properties, was essential in overcoming operational complexities and achieving the cementing objectives. This study highlights the importance of innovative solutions and a tunable engineering approach to achieve zonal isolation in complex and challenging cementing environments.

This paper presents insights into the design, execution, and evaluation of cementing operations through the effective use of innovative techniques. The novel integration of MPC, expandable liner hangers, and environmentally friendly lost circulation spacers contributed to ensuring well integrity. Furthermore, the paper offers valuable perspectives on how the integration of advanced methods and technologies can enhance operational excellence in cementing applications.

Introduction

Drilling oil and gas wells in the Middle East has long posed significant challenges due to the complex lithology of the region, as extensively documented in the literature (Turki, 1985). After successfully drilling through such formations, the next critical operation, primary cementing, plays a pivotal role in ensuring well integrity, production performance, and long-term reliability.

The primary objective of cementing extends beyond simply placing cement in the annular space between the casing or liner and the formation. Achieving zonal isolation requires meticulous planning, comprehensive design studies, advanced simulations, and extensive laboratory testing to engineer suitable cement systems and placement techniques.

Cementing across high-pressure formations intersecting weak zones presents one of the most demanding challenges in well construction, particularly when the operational window (the pressure range between pore and fracture pressures) is extremely narrow.

When encountering formations where high-pressure zones are intersected by weaker formations, the operational window can be as narrow as 0.5 lbm/ft³ in equivalent mud weight (Masoud et al., 2024; Quintero et al., 2024). This already presents a substantial challenge for cement placement, where the risk of either formation breakdown or fluid influx is elevated. The complexity is further intensified when operators choose to run a liner and set a liner-top packer within such a section. While the packer serves as a critical mechanical barrier to isolate the lower zone, it simultaneously introduces a significant reduction in the hydrostatic pressure exerted by the fluid column above it.

Once the liner-top packer is set, the annular cement system is exposed to formation pressures with limited support from above. This pressure imbalance can result in the unset cement slurry being displaced into the weak formation, particularly in the presence of narrow pressure margins. Such premature displacement may compromise zonal isolation and necessitate costly remedial operations, risking the loss of target zones or even the well itself, with severe financial implications.

Technology background

Managed pressure cementing

MPC is a technique designed to control the downhole equivalent circulating density (ECD) by regulating surface backpressure (SBP) during cementing operations. This is achieved through the use of a fully automated choke manifold system, which allows precise manipulation of annular pressure in real time.

The core equipment used in MPC includes:

- Rotating Control Device (RCD)
- Automated drilling-choke system

Rotating control device. The RCD is installed above the annular blowout preventer (BOP) and serves as a critical component of the MPC system. It creates a dynamic seal around the drill pipe, allowing closed-loop circulation by diverting the return fluids away from the rig floor and toward the choke manifold.

Key advantages of the RCD in MPC operations:

- Enables safe containment of annular fluids
- Supports string reciprocation and rotation during liner running and cementing
- Maintains constant bottomhole pressure (BHP) or equivalent mud weight (EMW) while running in hole (RIH)

This sealing and diversion capability is essential for maintaining pressure integrity during dynamic operations.

Automated drilling-choke system. The MPC choke manifold is a fully automated control system designed to manage SBP and maintain the downhole pressure within a defined EMW target.

Key functions and capabilities include:

- Real-time surface back pressure control to maintain BHP.
- Automatic detection of formation losses or influxes (kicks).
- Dynamic simulations to optimize SBP during cement placement.
- Auxiliary pump integration to instantly apply additional SBP in case of pump shutdown or unexpected flow interruption.

The automation and responsiveness of the MPC manifold allow it to proactively adapt to changes in well conditions, ensuring pressure consistency and cement placement integrity, especially in wells with narrow pore pressure and fracture pressure windows

The governing equation Eq.1 used in MPC is:

$$BHP = \text{Hydrostatic pressure} + AFL + SBP$$

Where:

BHP = Bottomhole pressure

AFL = Annular friction pressure

SBP = Surface Back Pressure

This approach enables the use of lighter-density fluids than those typically required in conventional overbalanced cementing. The MPC technique allows the EMW to be adjusted dynamically, maintaining wellbore pressure within the narrow operational window, between pore pressure and fracture gradient.

While MPC is not considered a routine cementing practice, it offers several advantages, particularly in high-risk or narrow operational window wells:

- Maintains constant ECD throughout the operation.
- Enables the use of lighter-weight drilling fluids.
- Allows the use of lighter density pre-flushes or spacers.
- Reduces risk of losses or influxes.
- Improves cement placement and zonal isolation in challenging formations.

MPC has proven especially effective in complex wellbore environments, where traditional cementing methods pose a high risk of inducing formation breakdown or experiencing losses.

Cementing fluids

Achieving reliable zonal isolation in complex well conditions, especially those involving narrow pressure margins, requires a meticulously engineered approach to cementing fluid design. The cement system must effectively withstand fluid influx, formation pressure variations, and potential losses, while maintaining placement integrity until it reaches full strength. In such high-risk environments, cement is not merely a sealing material but a barrier to isolate high-pressure formations from low-pressure zones, and to prevent formation fluid migration or sustained casing pressure.

Cementing fluid performance is governed by a number of interrelated factors including rheological behavior, density, compatibility with drilling fluids, fluid stability, and transition time. Each of these must

be carefully engineered and validated to ensure the slurry can be pumped under controlled conditions, maintain sufficient annular coverage, and develop the mechanical properties necessary for long-term sealing performance.

An integrated engineering cycle, spanning laboratory testing, displacement simulations, field execution, and post-job evaluation, is essential. This cycle not only ensures job success under current well conditions but also provides valuable feedback for continuous improvement in similar future operations. A holistic understanding of fluid interactions and downhole dynamics enables the tailoring of specific systems, such as wellbore shielding spacers and gas-tight cement slurries, that work in synergy with advanced placement techniques like MPC.

The following sections present in detail the design considerations and field validation of:

- Wellbore Shielding Spacer Design
- Cement Slurry Design
- Cementing Design and Simulations

Each of these components was engineered to work together as a unified system, ensuring operational success in wells with limited margins for error.

Wellbore shield spacer design. Cementing spacers play several critical roles in achieving a successful cementing operation. Their functions include effective displacement of the drilling mud from the annulus and casing/formation surfaces, separating the cement slurry from the drilling fluid, and ensuring compatibility with both fluids. Incompatible fluids can result in unpumpable mixtures. Additionally, spacers help ensure that annular surfaces are water-wet to promote cement bonding.

Cementing spacers are typically composed of polymers, weighting agents, surfactants, and solvents, particularly when used to remove non-aqueous drilling fluids (NAF). Spacers must be designed to have a density higher than that of the mud and lower than that of the cement (density hierarchy). They must also exhibit higher shear stress than the mud but lower than the cement (rheology hierarchy).

The flow regime is another essential criterion in spacer design. Optimized mud removal requires uniform annular velocity. While **turbulent flow** is ideal, it is often impractical due to the high pumping rates required and the associated frictional pressures that could induce formation losses, reduce top of cement, and result in poor cement coverage. Therefore, **effective laminar flow** is usually the target. To achieve this, four key criteria must be met:

- At least 10% density hierarchy,
- 20% friction pressure hierarchy,
- Adequate casing stand-off to ensure fluid flow around the casing, and
- Uniform annular fluid velocity.

In challenging formations with narrow operational windows, these criteria may not be achievable, especially when used in combination with resilient NAF drilling fluids. This complicates spacer design and increases the risk of fluid losses during cementing. To address these challenges, a **novel wellbore shielding spacer** has been developed. This spacer meets all conventional requirements and contains flexible solid particles of various sizes and shapes. These particles form an impermeable barrier against the formation, reducing pressure transmission, expanding the operational window, and enabling higher ECDs without compromising wellbore integrity.

The plugging particles in the spacer are activated upon exiting the float shoe and contacting the formation. These particles position themselves on the formation face without invading the matrix and are later removed

during underbalanced conditions in well cleaning, testing, and production phases, minimizing formation damage. (See Figure 1.)

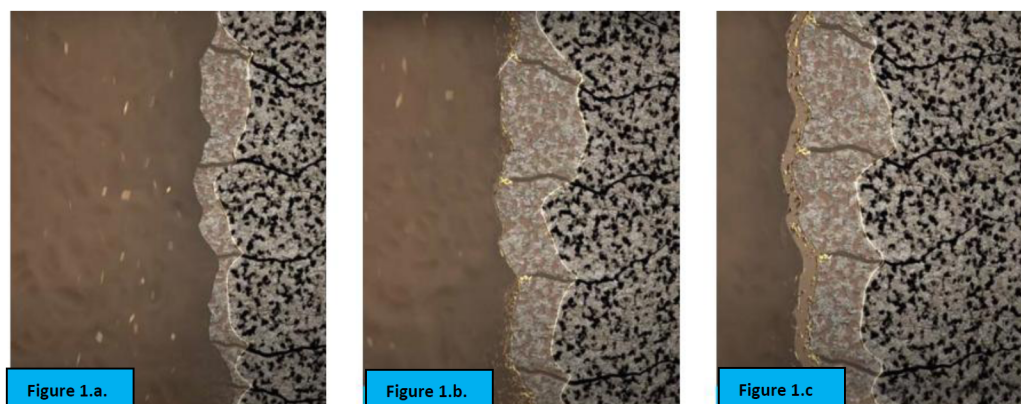


Figure 1—wellbore shielding particles dispersed within the spacer (1.a). The plugging particles start to accumulate in the formation pore throats (1.b). An impermeable barrier is created in the formation face with minimum penetration, stopping the pressure transmission and fluid invasion into the formation (1.c)

The wellbore shielding spacer can plug pore throats up to 250 microns. When combined with a naturally occurring, environmentally friendly LCM with a broader particle size distribution (PSD), plugging capabilities can be extended up to 1,000 microns or 3,500 darcies. This combination provides differential pressure resistance exceeding 1,000 psi at maximum sealing capacity.

These particles, while plugging pore throats, also aid in removing difficult-to-remove gelled mud that compromises cement bonding and may lead to issues such as microannuli, hydrocarbon migration, and sustained casing pressure. The wellbore shield spacer also demonstrates excellent fluid compatibility, maintaining acceptable ECDs and frictional hierarchy during cementing.

The wellbore shielding spacer is suitable for use at temperatures up to 350°F (176°C) and has stable rheological properties under downhole conditions to ensure proper solid dispersion. This is especially critical in liner cementing operations, where narrow clearances and high friction pressures are common.

Unstable fluids may result in solids sagging or agglomeration, which could clog surface or downhole restrictions. However, this spacer system, used at concentrations of 8–20 lbm/bbl, combined with 20–25 lbm/bbl environmentally friendly LCM has been successfully deployed in multiple field operations without any service quality incidents. Proper fluid design, with effective solids dispersion and appropriate surface and downhole rheology, is key to operational success.

Table 1 shows a database of liner hangers where the wellbore shield spacer system was successfully deployed. Two parameters are essential for analysis: **annular total flow area (TFA)** and **linear clearance**. TFA significantly influences friction pressure, smaller TFA leads to higher pressure drops. For instance, the 4½-in liner hangers from Companies B and C show the smallest TFA (1.73 and 1.72 in², respectively). Linear clearance, the difference between the previous casing ID and liner hanger OD, also affects the design. The 7-in liner hanger from Company B shows the smallest clearance (4,064 microns), which exceeds the maximum plugging capacity of the spacer-LCM system, eliminating any potential undesired plugging, and based on an optimized spacer shield design.

Table 1—Liner hanger's database from different manufacturers where the wellbore shielding spacer paired with the 1,000 microns LCM has been successfully deployed

Liner Hanger Company	Liner Size (in)	Model or Type	Element	Previous Csg ID (in)	OD (in)	Linear Clearance (in)	Linear clearance (microns)	ID (in)	Length (ft)	TFA (in ²)
Company A	7	7-in × 9-5/8-in Balanced Hydraulic Liner Hanger	7-in 32# Tie Back Receptacle	8.535	8.25	0.285	7,239	7.375	17	3.76
			7-in 32# Liner Top Packer	8.535	8.226	0.309	7,848.6	6.17	4	4.07
			7-in 32# Balanced Rotating Liner Hanger	8.535	8.23	0.305	7,747	6.17	11	4.02
Company B	7	7-in × 9-5/8-in expandable Liner Hanger	Expandable Liner Hanger QUICK-LOCK LINER HANGER (Seal Bore)	8.535	8.375	0.16	4,064	7.25	4.45	2.12
			Expandable Liner Hanger QUICK-LOCK LINER HANGER TYPE	8.535	8.375	0.16	4,064	6.509	7.62	2.12
			Reducing Adapter	8.535	8.029	0.506	12,852.4	6.064	3.1	6.58
Company C	7	Liner 7-in × 9.625-in	Liner top packer - TBR 7-in x 9 5/8-in	8.535	8.25	0.285	7,239	7.375	15	3.76
			Company C w/Anti-Preset Feature liner top packer	8.535	8.312	0.223	5,664.2	6.025	7.15	2.95
			7-in × 9-5/8-in Control Set Flex Lock 5 Liner Hanger	8.535	8.234	0.301	7,645.4	6.025	14.2	3.96
Company C	4.5	4 1/2-in × 7-in Liner Flex	26 ft extension	6.004	5.75	0.254	6,451.6	3.2	26	2.34
			Top Packer	6.004	5.77	0.234	5,943.6	2.84	5.45	2.16
			4.5 × 7-in Control Set FL Liner	6.004	5.819	0.185	4,699	2.6	9.8	1.72
Company D	7	7-in × 9 5/8-in Liner Packer STD	8,310-in x 7,500-in SA-6 P WPBR	8.681	8.31	0.371	9,423.4	7.5	11.41	4.95

Liner Hanger Company	Liner Size (in)	Model or Type	Element	Previous Csg ID (in)	OD (in)	Linear Clearance (in)	Linear clearance (microns)	ID (in)	Length (ft)	TFA (in ²)
			7-in x 9.625-in HNGR-PCKR	8.681	8.334	0.347	8,813.8	6.187	8.5	4.64
Company C	9.625	13.375-in × 9 5/8-in Liner Packer	9-5/8-in TBR 10.188-in Ext.ID 11.68-in Ext.OD 20ft Ext.LG	12.347	11.68	0.667	16,941.8	10.188	20	12.59
			9-5/8-in ZXP Liner Top Packer	12.347	11.7	0.647	16,433.8	8.4	6	12.22
			9-5/8-in 58.4# ControlSet Hydraulic Hanger, Flex 5 NON ROT	12.347	11.718	0.629	15,976.6	8.4	20	11.89
Company B	4.5	4 1/2-in × 7-in Expandable Liner Hanger	Expandable Liner Hanger QUICK-LOCK LINER HANGER TYPE (Seal Bore)	6.094	5.91	0.184	4,673.6	4.999	3.2	1.73
			Expandable Liner Hanger QUICK-LOCK LINER HANGER TYPE (Liner Hanger)	6.094	5.91	0.184	4,673.6	4.52	9.88	1.73
			Reducing Adapter	6.094	5.662	0.432	10,972.8	3.838	3.44	3.99
Company C	4.5	5-in × 7-in Liner Flex	Liner Top Packer	6.276	5.818	0.458	11,633.2	3.2	26	4.35
			w/Anti-Preset Feature liner top packer	6.276	5.874	0.402	10,210.8	2.84	5.45	3.84
			5 × 7-in Control Flex Lock	6.276	5.978	0.298	7,569.2	2.6	9.8	2.87

To ensure field suitability, all necessary laboratory tests are conducted according to API RP B2, including evaluations of rheology (surface and downhole), fluid compatibility, and wettability. Additional tests verify plugging capacity and stability.

Plugging capacity is tested using a **Particle Plugging Apparatus (PPA)**. (See [Figure 2](#) and sample slot images ranging from 1,000 to 3,000 microns.) The fluid is subjected to a sudden 1,000 psi differential pressure across a test slot. Initial spurt loss and total fluid loss are measured in grams to assess efficiency.

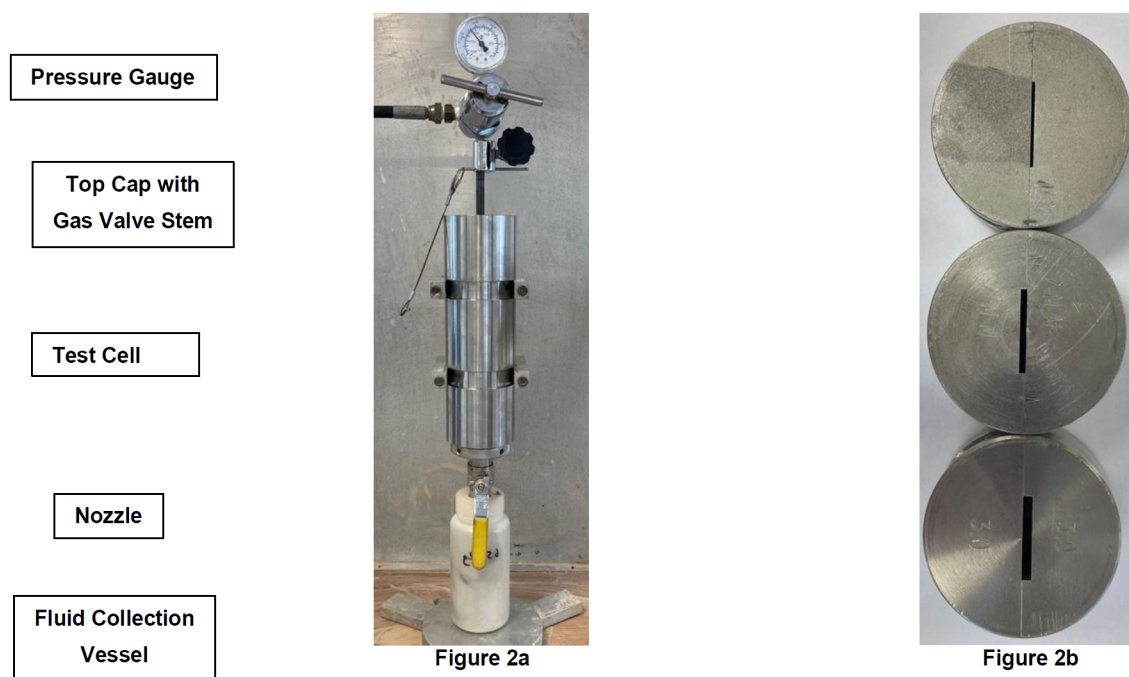


Figure 2—Particle plugging apparatus (PPA) and sample slots

For both case studies the slot test was performed successfully. In [Table 2](#) the formulation of the 142 lbm/ft³ wellbore shielding spacer paired the 25 lbm/bbl LCM, 1,000 psi differential pressure to the spacer sample with 17% volume of fluid loss in the initial spurt. Pressure was maintained for 30 minutes to ensure the durability of the formed shield. The 30 total fluid losses accounted for 35% of the total fluid volume sample. The 1,000 microns slot after the test is shown in [Figure 3](#).

Table 2—142 lbm/ft³ wellbore shielding spacer formulation for case study 1

Description	Concentration	Unit	Lot number
Base fluid	0.502	bbbl/bbl	Rig sample
Antifoam	0.2	gal/bbl	Laboratory sample
Wellbore shielding spacer	9	lbm/bbl	Laboratory sample
Environmentally friendly LCM	25	lbm/bbl	Laboratory sample
Weighting agent	547	lbm/bbl	Laboratory sample
Surfactant	3	gal/bbl	Laboratory sample
Mutual solvent	2	gal/bbl	Laboratory sample

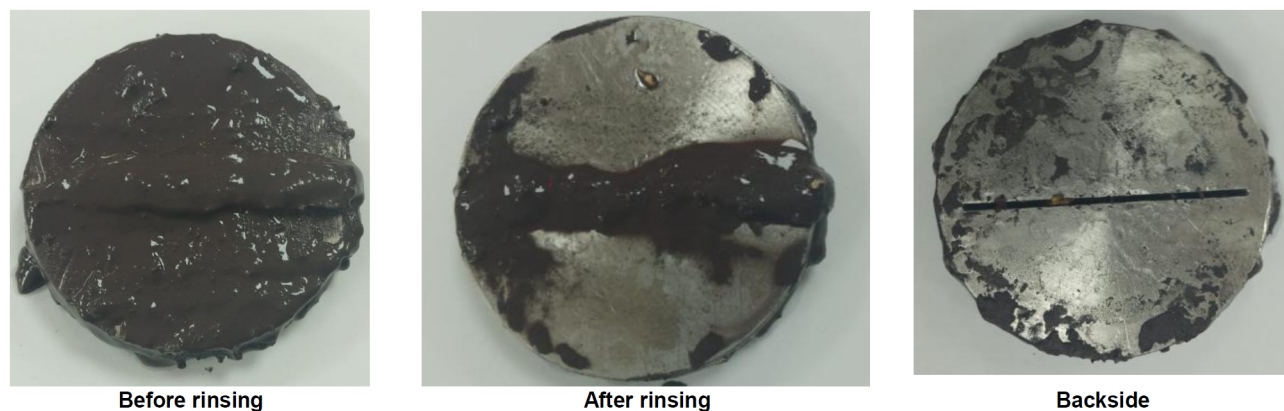


Figure 3—142 lbm/ft³ - 1,000 microns slot test at 1000 psi differential pressure for case study 1

To evaluate downhole stability, fluid samples are placed in a **High-Pressure High-Temperature (HPHT) consistometer**. The system simulates downhole conditions for 3–4 hours (1 hour for GO/NO-GO liner applications), and the chart is evaluated for any fluid degradation (Figure 4). Post-test, the paddle, cup, and base are visually inspected. Density is measured at the top and bottom to identify variation. (See Figure 5)

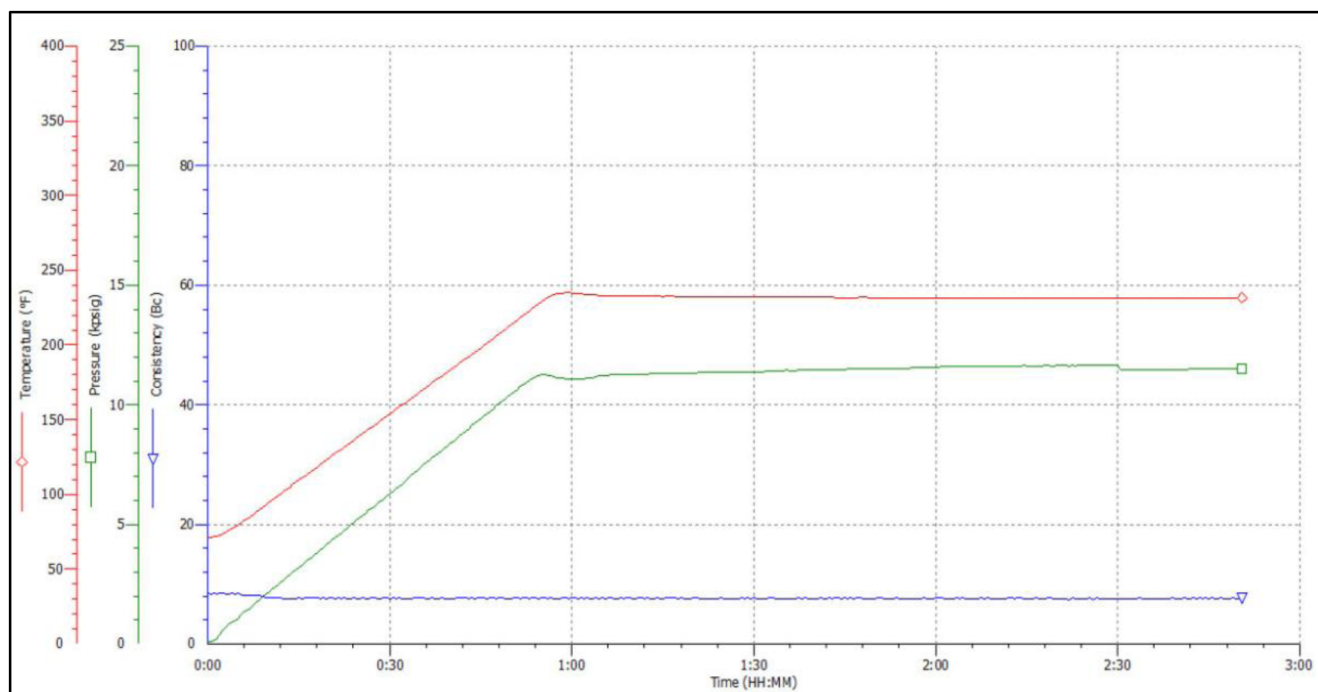


Figure 4—HP/ HT consistometer chart of wellbore shield spacer under a field application downhole condition



Figure 5—Visual inspection of the HPHT consistometer cell, paddle and bottom cap for the wellbore shield spacer of case study 1.

For further validation, the same fluid is poured into a 250 ml graduated cylinder at a 45° angle and left under static conditions for two hours. Density is measured at the top, middle, and bottom using a syringe. In **case study 1**, the design density was 142 lbm/ft³, with measured values of 141.5 (top), 142.4 (middle), and 142.8 (bottom), showing less than a 1 lbm/ft³ variation, confirming fluid stability. (See [Figure 6](#))



Figure 6—Static stability test for the wellbore shield spacer used in case study 1

Cement slurry design. In addition to high-performance spacer, precisely designing slurries to address potential risks is crucial for the presented cases. While considering the cement placement complications, such as longer static period to set the liner top packer and efficient mud removal without exceeding fracture limitations, providing required zonal isolation post cement placement must be achieved as well.

For case-2, to account for the contingency plan of setting and expanding the hydraulic liner hanger packer if its main setting technique was unsuccessful, the slurry across the top of the liner should withstand a 120-minutes static condition. Achieving the required thickening time with this relatively long static condition without severely affecting other properties is challenging, especially with higher densities' slurries. While following API testing procedures and considering the operator's best practices, the slurry in case-2 was tested with a two-hours stoppage time to account for the liner top packer expansion as in [Figure 7](#) below:

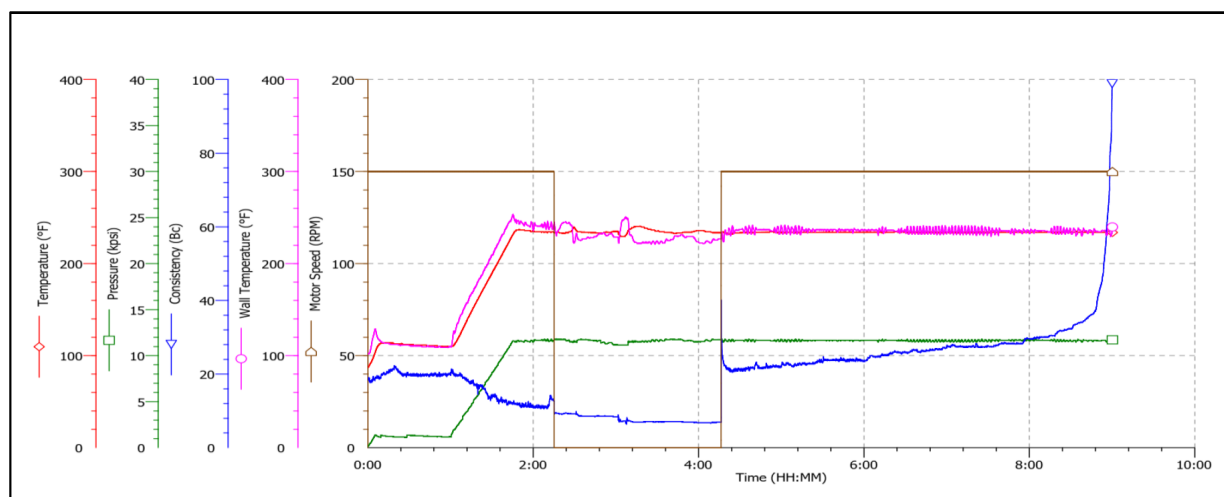


Figure 7—TT of case-2 slurry with 2 hours static condition

The properties of the cement slurries across the high-pressure formation post cementing are crucial to ensure secure execution of the cement job. One of these main properties is the slurry's capability of preventing gas flow during critical gel time. Gas block additives were used in the design, and a static gel strength test was conducted to slurries across the critical zone to ensure they can restrict gas migration in that period. Setting a liner top packer, despite creating sustainable mechanical separation above the formation, would isolate all the hydrostatic pressure associated with the fluids above it and any back pressure applied by MPC to control formation pressure, hence increasing the possibility of gas flow if slurry is not able to prevent it. The Figure 8 below illustrates the static gel strength test conducted for slurry in of both cases, where the transition time between 100 lbf/100 ft² to 500 lbf/100 ft² is less than 40 minutes

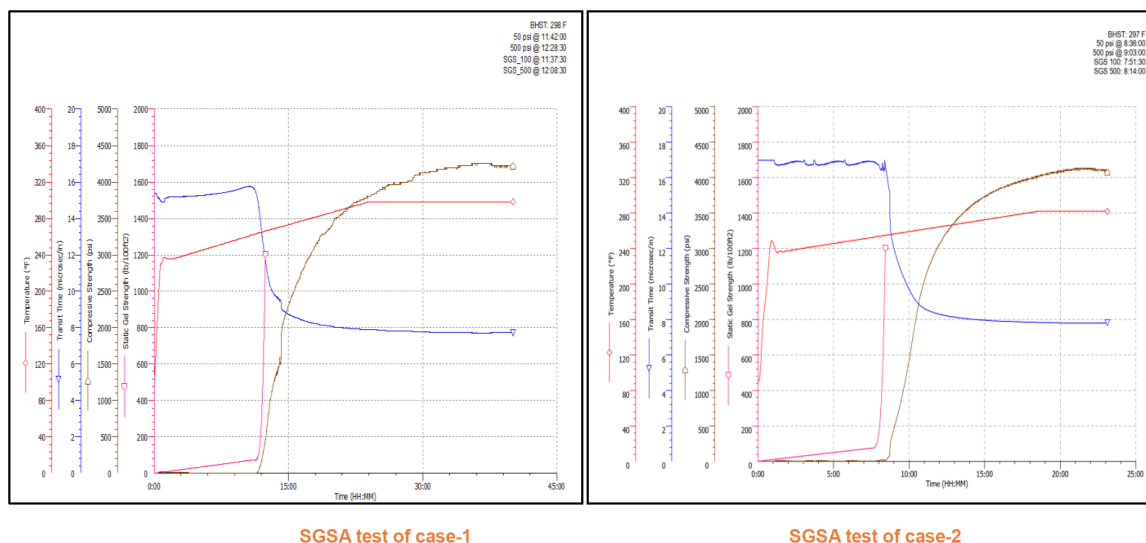
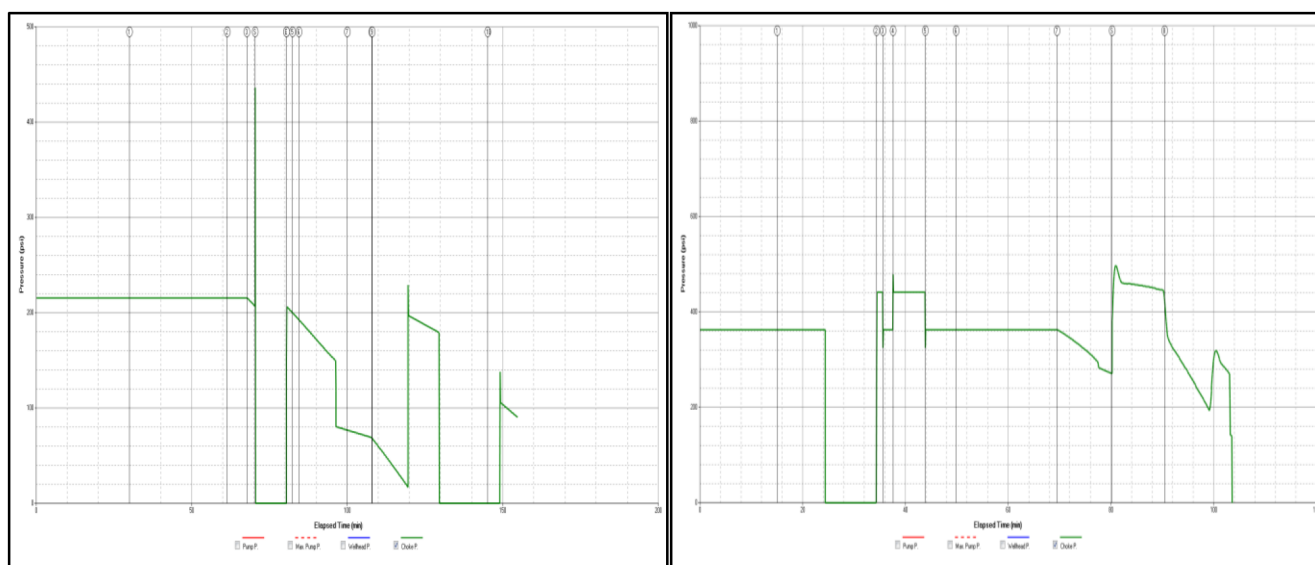


Figure 8—Static gel strength test for slurries across high pressure zones for both cases

Adhering to the API testing procedures, the distribution of solid particles and the slurry stability tests were conducted for both cases. Nonhomogeneous cement columns in these critical sections can severely jeopardize zonal isolation. Another crucial aspect to be addressed is the micro-anulus risk, which can occur due to the bulk shrinkage of the cement system. Expansion additives were integrated into both cases' cement system design to mitigate this risk, and an expansion test was conducted as per API-10B5.

Prior to initiating the blending of additives, the specific gravity of each blend component was measured using a pycnometer. To ensure the blend was homogeneous, the blending process was carried out by sandwich procedure in addition to several transfers between silos. Samples from each portion of the blend were collected, and their specific gravity was compared against the theoretical specific gravity before mobilizing to location.

Cementing design and simulations. Aligned with cementing fluids' design, studying the well and condition of each case is a crucial aspect of a successful cementing operation. Extensive simulations of the back pressure to be applied for MPC operations were conducted by studying the fluids' properties, pumping rate, and pumping pressure to ensure safe placement and maintain the ECD above the pore pressure and below the frac pressure while cementing. [Figure 9](#) below reflects the pre-job back pressure prediction for both cases.



Surface back pressure simulation for case-1

Surface back pressure simulation for case-2

Figure 9—Surface back pressure simulation for both cases

An additional key factor to be considered during pre-job planning is the mud removal, despite having optimum cement slurry properties, mud contaminating the slurry can jeopardize the compressive strength development of a set cement, resulting in zonal isolation failure. Completely removing mud from the annulus to be cemented requires attention to pipe centralization. However, open hole caliper log could not be conducted for both cases. This made knowing the washout zones inaccessible, which added another challenge to achieving the required stand-off profile. Comprehensive simulations were conducted with several considerations from gauged hole to an annular excess of 50% in case-1 and annular excess of 30% in case-2 to agree on the centralization plan, as can be seen in [Figure 10](#) below.

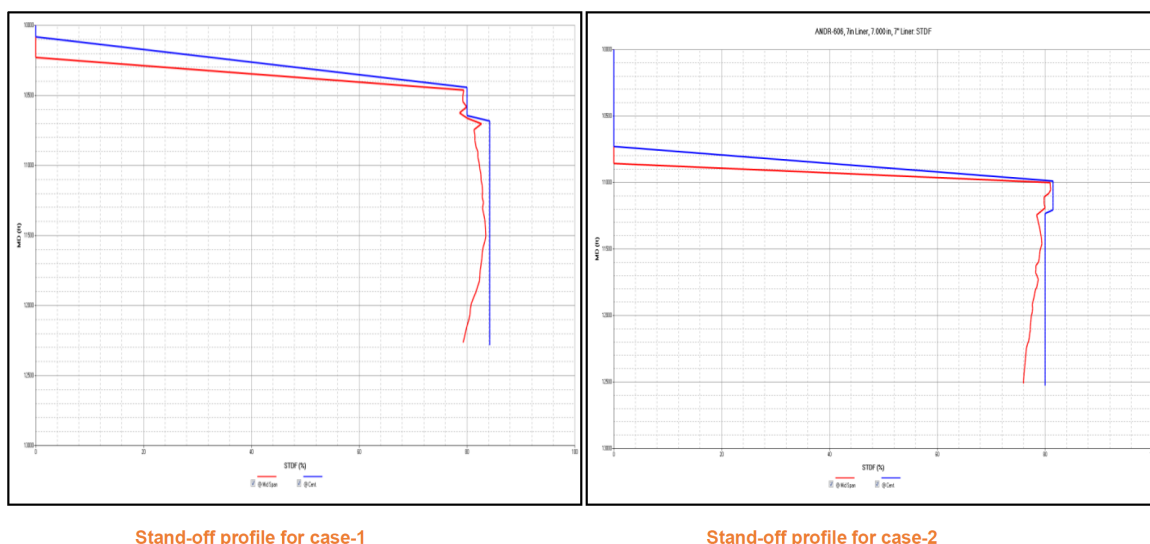


Figure 10—Centralization simulation for both cases

The result of integrating average stand-off profiles higher than 70% and enhanced fluids friction pressure hierarchy showed a promising displacement efficiency simulation during the planning phase. Figure 11 below shows the mud removal simulation for case-1 at the left with 50% excess over 12-in OH and case-2 with 30% excess over 8 3/8-in OH.

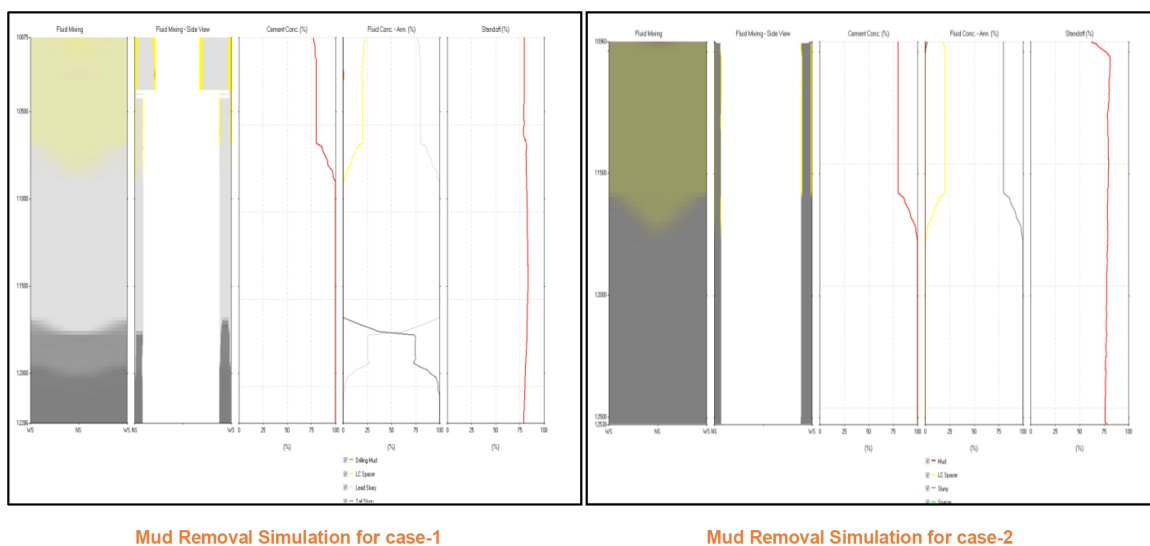


Figure 11—Displacement efficiency simulation during planning

The cement across the TOL was predicted to be contaminated with at least 20% spacer in both cases, which was proven to develop sufficient compressive strength as per the UCA charts shown below in Figure 12.

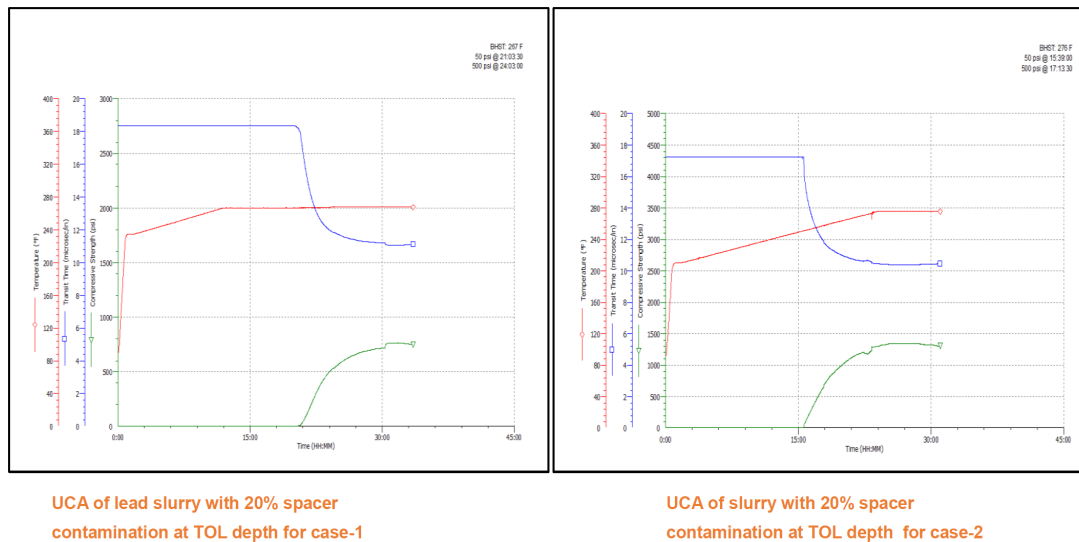


Figure 12—UCA charts of slurries with 20% spacer contamination at TOL condition

Cases studies

Case study #1

This case study involves a gas development well where a deviated 12-in OH section was drilled from the previous 13 3/8-in casing shoe using 139 lbm/ft³ OBM using MPD equipment. The MPD system maintained a constant BHP equivalent to 145 lbm/ft³ EMW. Formation integrity tests indicated a pore pressure of 144 lbm/ft³, while losses were observed at 145.6 lbm/ft³ EMW, resulting in a very narrow operational window of only 1.6 lb/ft³. Drilling progressed under constant partial losses.

After running the 9 5/8-in liner to planned TD and setting the liner hanger, the well was circulated at 5 bpm, with observed losses of 20 barrels per hour (BPH). The primary cementing operation was executed using the MPC system to maintain 145 lbm/ft³ EMW. The cementing sequence included:

- 150 bbl of 142 lbm/ft³ wellbore shielding spacer, mixed with 25 lbm/bbl of environmentally friendly LCM
- 158 bbl of 145 lbm/ft³ engineered lead cement slurry, batch mixed
- 46 bbl of 150 lbm/ft³ engineered tail cement slurry, batch mixed
- Drop of a liner dart
- 10 bbl of additional tail cement slurry
- 10 bbl of 145 lbm/ft³ conventional spacer
- Displacement with 145 lbm/ft³ OBM (kill mud)

During displacement, partial losses ceased as the wellbore shielding spacer passed through the loss circulation zone (LCZ). Clear signs of successful dart latching were observed, and the liner wiper plug was bumped at the landing collar as expected. The float equipment functioned properly, and the liner top packer was set. The running tool was released, 12 stands were pulled out of hole (POOH), and circulation was initiated using overbalanced OBM. A total of 40 bbl of pure spacer and 30 bbl of contaminated spacer were recovered at surface.

Following the waiting on cement (WOC) period, the rig ran in hole (RIH) and tagged the TOC approximately 300 ft above the TOL. Hard cement was drilled through the TOL and the shoe track. Both positive and negative pressure tests confirmed zonal isolation

Pressure Match Analysis. As shown in Figure 13, the comparison between actual surface pump pressures (blue line) and simulated pressures (red line) demonstrates close matching throughout the job. The simulation revealed an OH size of 12-in with 50% annular excess. Clear pressure spikes corresponding to dart latching and plug bumping confirm proper execution. While the use of higher-density OBM (145 lbm/ft³) during displacement covered losses detection in pressure match analysis, field observations verified that losses stopped once the shielding spacer crossed the LCZ, and TOC was found as expected above the liner top.

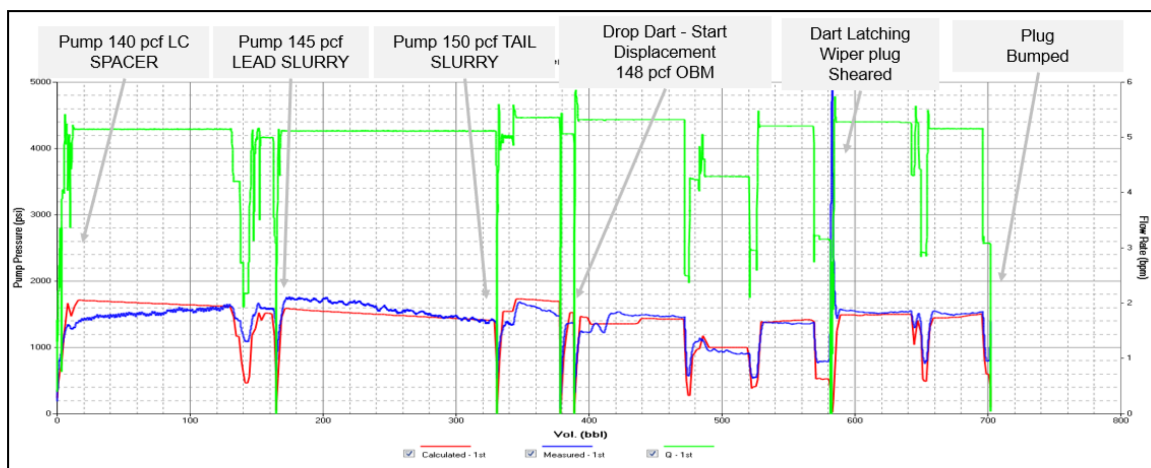


Figure 13—Post cementing pressure match analysis for case study #1. Blue Line represents the actual pumping pressure and red line represents the simulated pumping pressure.

ECD Control Analysis. The ECD profile shown in Figure 14 confirms that ECD was maintained within the narrow operational window throughout the cement job. Minor excursions slightly above 146 lbm/ft³ occurred momentarily but remained within acceptable operational limits.

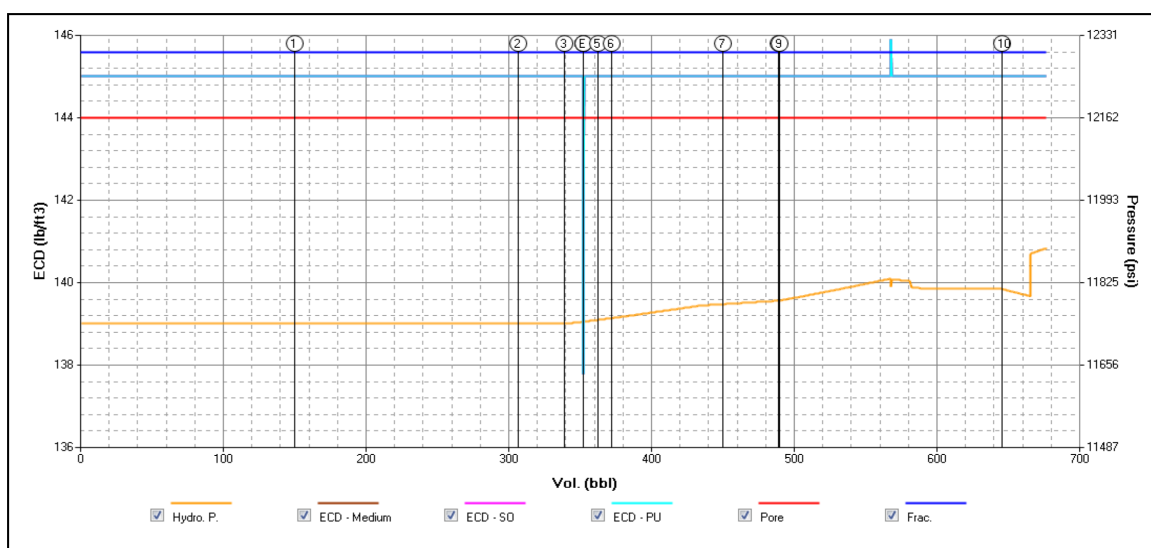


Figure 14—Post cementing ECD analysis for case study #1.

Case study #2

This case study focuses on a re-entry well where a window was milled from the existing 9 5/8-in. casing, and an 8 3/8-in. deviated open hole section was drilled using 135 lbm/ft³ OBM under MPD technique. The MPD system was employed to maintain a constant BHP equivalent to 144 lbm/ft³ EMW. A drop of just 1

lbm/ft³ in EMW (to 143 lbm/ft³) resulted in wellbore influx and mud weight reduction to 134 lbm/ft³. The fracture pressure for the section was estimated at 150 lbm/ft³, indicating a very narrow pressure window.

The 7-in liner was successfully run to total depth. Prior to cementing, two full bottom-up circulations were performed to clean the hole and reduce drilling fluid rheologies. The primary cementing was executed using the MPC system, maintaining 144 lbm/ft³ EMW, and the sequence consisted of:

- 75 bbl of 140 lbm/ft³ wellbore shielding spacer, blended with 25 lbm/bbl of environmentally friendly LCM
- 53 bbl of 145 lbm/ft³ engineered cement slurry, batch mixed
- Drop of liner dart
- 5 bbl of additional cement slurry
- 10 bbl of 140 lbm/ft³ conventional spacer
- Displacement with 148 lbm/ft³ OBM (kill mud)

Indications of successful dart latching were observed, and the liner wiper plug was bumped at the landing collar as expected. Additional pressure was applied to set and expand the liner top packer under static conditions. The running tool was then released, and reverse circulation was initiated using overbalanced OBM. Contaminated spacer and 10 bbl of clean cement slurry were recovered at the surface, evidence of a high-quality cement placement reaching the TOL.

Following the WOC period, the rig successfully drilled hard cement in the shoe-track, and performed a positive and negative tests to confirm the zonal isolation.

Pressure Match Analysis. As shown in Figure 15, the pressure match between actual surface pump pressures (blue line) and simulated pressures (red line) demonstrates excellent alignment throughout the job, when considering 8 3/8-in OH with 30% OH excess. Clear indications of dart latching and plug bumping at the end of displacement confirm proper execution. Due to the use of higher-density or kill OBM (148 lbm/ft³) during displacement, identification of losses based on the pumping pressure was not feasible; however, no losses were reported by the field team, and the expected cement volumes were circulated to surface.

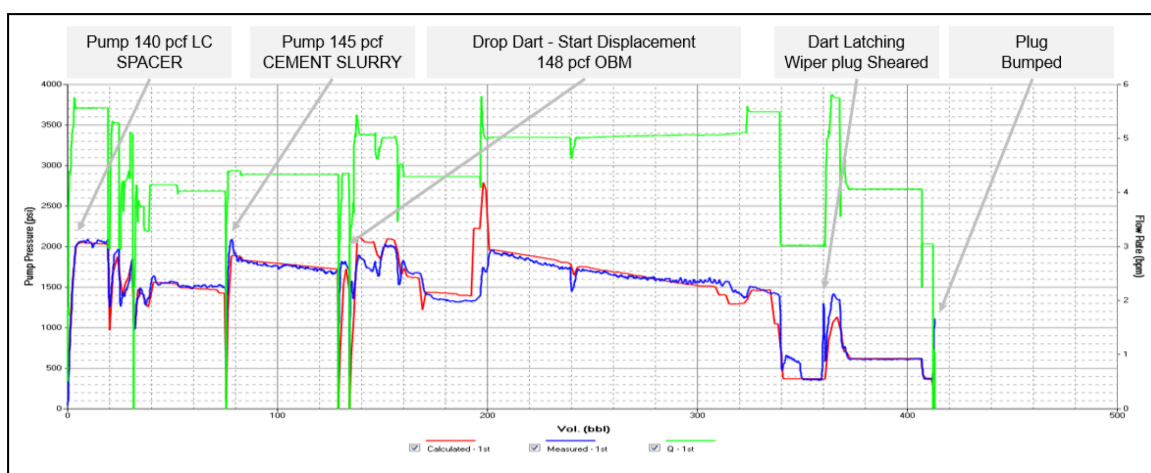


Figure 15—Post cementing pressure match analysis for case study #2. Blue Line represent the actual pumping pressure and red line represent the simulated pumping pressure.

ECD Control Analysis. The ECD profile (Figure 16) shows that the cementing operation maintained ECD within the narrow operational window, with a maximum ECD of 150 lbm/ft³, equal to the fracture gradient but without inducing losses.

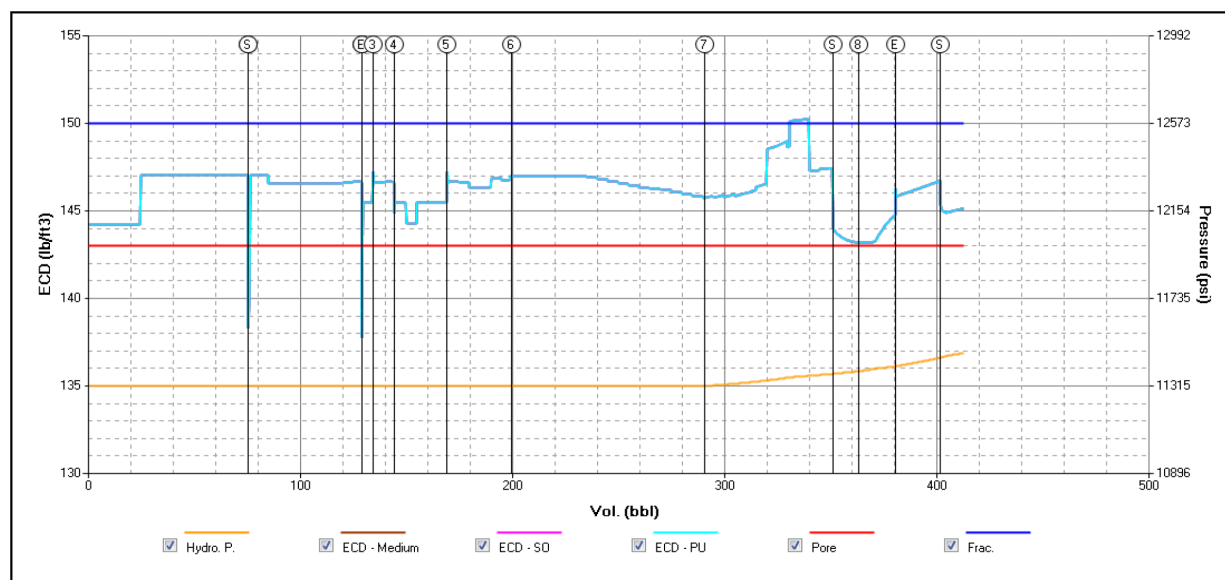


Figure 16—Post cementing ECD analysis for case study #2.

Notably, in both cases, when the liner top packer was set, considerable part of hydrostatic pressure was cut off, significantly reducing the annular pressure below the packer. This pressure drop meant that the confined annular pressure became lower than the formation pore pressure, creating a risk of fluid influx or migration, which could have displaced unset cement slurry into the loss zone, thereby compromising zonal isolation. The effectiveness of the wellbore shielding spacer system played a crucial role in preventing such losses, allowing the gas-tight cement to set undisturbed and ensuring the long-term integrity of the isolated section.

Conclusion

The combination of advanced techniques and a robust engineering approach presented in this paper can be effectively replicated in wells facing similarly narrow operational windows. Meticulous control of downhole ECD, the application of a wellbore shielding spacer, and the use of an engineered cement slurry system were all essential to achieving successful zonal isolation in these critical well scenarios. The case studies demonstrate that a tailored and integrated cementing strategy is key to overcoming the complex challenges encountered in high-pressure, weak formation environments.

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Nomenclature

<i>AFL</i>	= Annulus Friction Pressure
<i>API</i>	= American Petroleum Institute
<i>BHA</i>	= Bottomhole Assembly
<i>BHP</i>	= Bottomhole Pressure
<i>CBHP</i>	= Constant Bottomhole Pressure
<i>COA</i>	= Certificate of Analysis
<i>CSGS</i>	= Critical Static Gel Strength
<i>ECD</i>	= Equivalent Circulating Density

<i>EMW</i>	= Equivalent Mud Weight
<i>FG</i>	= Fracture Pressure
<i>FIT</i>	= Formation Integrity Test
<i>HPHT</i>	= High-Pressure, High-Temperature
<i>KPI</i>	= Key Performance Indicators
<i>LCM</i>	= Loss Circulation Material
<i>MD</i>	= Measured Depth
<i>MPC</i>	= Managed Pressure Cementing
<i>MPD</i>	= Managed Pressure Drilling
<i>OH</i>	= Open Hole
<i>OBM</i>	= Oil-Based Drilling Fluid
<i>PSI</i>	= pounds per square inch
<i>QA/QC</i>	= Quality Assurance / Quality Control
<i>RCD</i>	= Rotating control device
<i>RIH</i>	= Run in Hole
<i>SBP</i>	= Surface Back Pressure
<i>SCP</i>	= Sustained casing Pressure
<i>SPP</i>	= Surface Pumping Pressure
<i>SGSA</i>	= Static Gel Strength Analyzer
<i>TD</i>	= Total Depth
<i>TOC</i>	= Top of Cement
<i>TOL</i>	= Top of Liner
<i>TTT</i>	=Thickening Time Test
<i>TVD</i>	= True Vertical Depth
<i>WOC</i>	= Waiting on Cement

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